

Name: _____

CIS 351 Sample CL3 Problem

1 September 2025

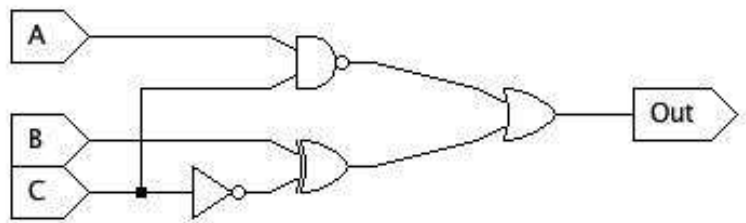
CL3: Circuit Representation

(a) Draw a circuit that implements the truth table below:

A	B	C	Out
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

(b) Complete the truth table below to show the output of the circuit for each input:

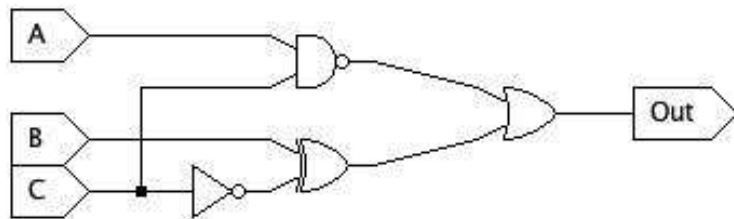
A	B	C	Out
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	



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(c) Draw a combinational circuit that implements this Boolean expression: $\overline{(AC)} + (B \oplus \bar{C})$

(d) Write the Boolean expression that implements the following circuit:



CIS 351 Sample CL3 Problem Solutions

Mon 21st Sept, 2026

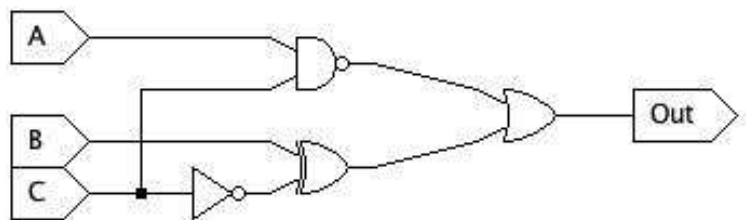
CL3: Circuit Representation

(a) Draw a circuit that implements the truth table below:

A	B	C	Out
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

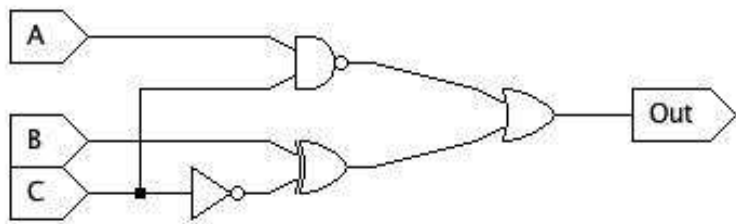
(b) Complete the truth table below to show the output of the circuit for each input:

A	B	C	Out
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1



(c) Draw a combinational circuit that implements this Boolean expression: $\overline{(AC)} + (B \oplus \bar{C})$

(d) Write the Boolean expression that implements the following circuit:



$$\overline{(AC)} + (B \oplus \bar{C})$$